

FSOC Coarse Pointing, Acquisition & Tracking - Performance Report

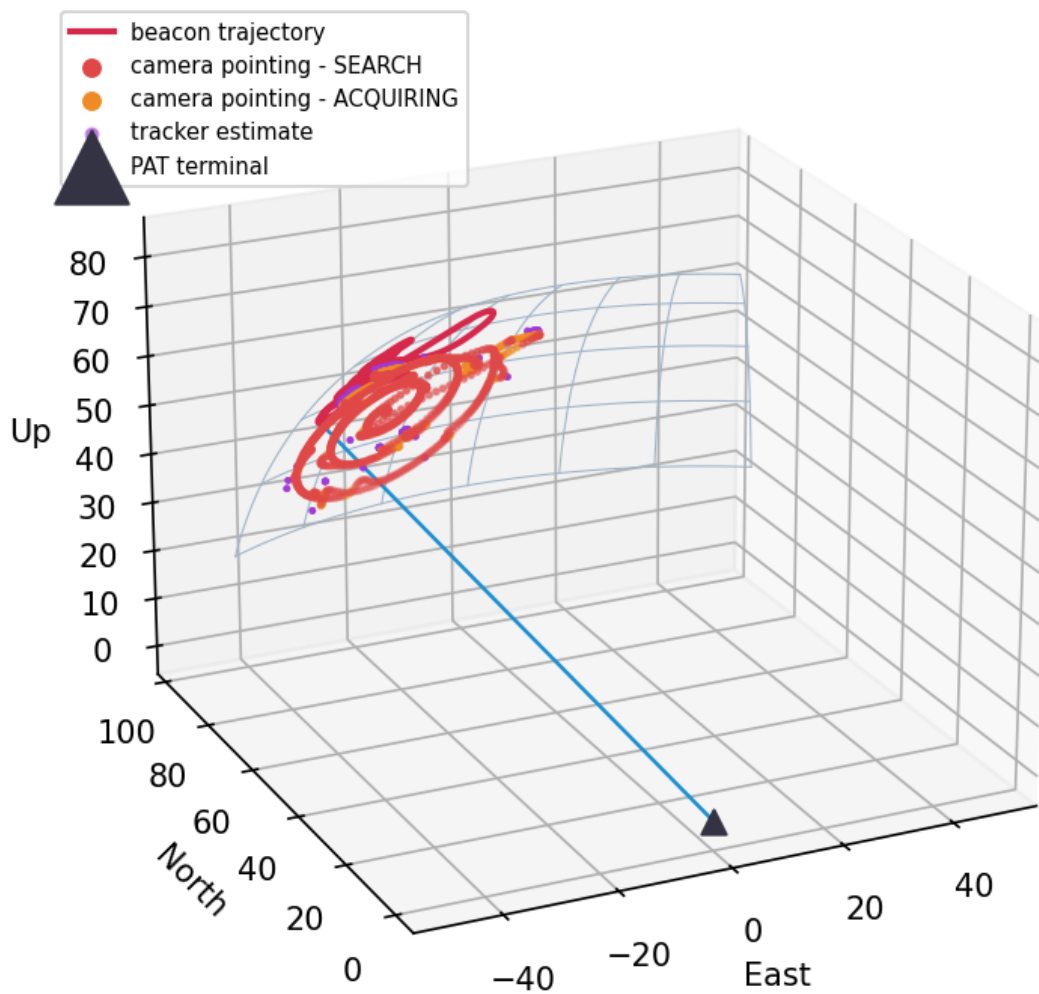
Run	
Generated	2026-09-29T21:50:01
Seed	1291733528
Identification method	hybrid
Beacon path	random_walk at 4.6 deg/s
Decoys	3 (2 blinking)
Turbulence / vibration	0.61 / 1.93
Clouds	yes
Initial pointing error	5.72 deg

Performance summary

Metric	Value
Simulation duration	41.4 s
Frames simulated	1242
Achieved frame rate	21.7 FPS
Processing time (mean / max)	2.88 / 5.52 ms
Processing throughput	347 FPS
Acquisition time	not acquired
Time locked on beacon (whole run)	0.0 %
Lock retention (after first lock)	0.0 %
Lock retention while beacon visible	0.0 %
Lock losses	0
Tracking error mean / RMS / max	- / - / - px
Tracking error mean / max	- / - mrad
Ready for fine-pointing handover	-
False-lock frames	0
Candidates rejected / locks dropped	3 / 0
Beacon in field of view / occluded	11.2 / 10.0 %

3D view of the run

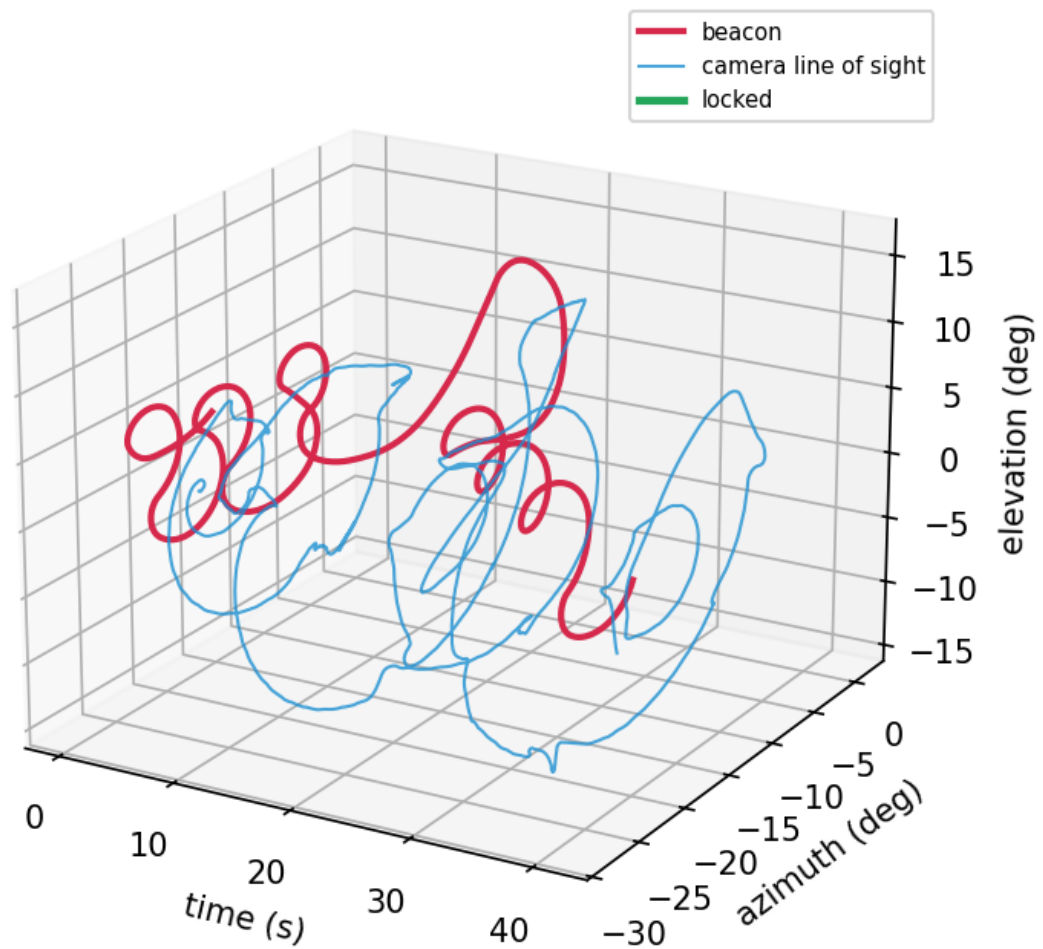
3D view: beacon trajectory and camera line of sight on the sky dome



Red: true beacon path. Dots: where the camera pointed, coloured by tracker state (red search, orange acquiring, green tracking, yellow coasting). Purple: tracker estimate.

3D view: azimuth / elevation over time

3D view: azimuth / elevation over time - beacon vs camera pointing



The camera line of sight (blue, green while locked) converging onto and following the beacon (red) shows acquisition and tracking in one picture.

Time charts

